Unified Cauer Ladder + Surface Impedance Network via Turning-Point Basis Switch

K. Sugahara*, H. Nagamine†, and Y. Hane‡

*Faculty of Science and Engineering, Kindai University, 3-4-1 Kowakae, Higashi-Osaka, Osaka 577-8502, Japan
†Faculty of Engineering, Gifu University, 1-1 Yanagido, Gifu 501-1193, Japan
‡Faculty of Science and Engineering, Toyo University, Kawagoe, Saitama, Japan
E-mail: ksugahar@kindai.ac.jp

***Abstract*—A truncated Cauer Ladder Network (CLN) is seamed to a Surface Impedance Boundary Condition (SIBC) element ($\alpha = 1/2$ Constant Phase Element) at a turning point ω_N given by Matsuo's corner-frequency formula, with the SIBC scaling fixed by continuity (no fitting). Formally this is the s_∞ expansion point of the multiple-expansion-points CLN framework. Analytical verification on a Cu slab reproduces the Stoll/Wall closed form across nine decades with ten Cauer rungs and one SIBC element; on a 3D Cu cuboid, an mpmath single-element VIM with HDiv divergence-free hex basis and an NGSolve HCurl Tanimoto-AT extraction yield matching unified $Y(j\omega)$ from two-to-three bulk Cauer rungs plus the SIBC element, cross-validated against a direct frequency-domain AC sweep.**

***Index Terms*—Cauer Ladder Network, Surface Impedance, Turning Point, Expansion Points, Eddy Current.**

I. Unified CLN + SIBC

A truncated Cauer Ladder Network (CLN) [1] represents the eddy-current admittance $Y(s) = 1/Z(s)$ as a continued fraction with rungs (R_{2k}, L_{2k+1}) encoding the Foster eddy-mode spectrum. The reliable rung count is bounded both by Hankel-Padé ill-conditioning ($\sim 300\times$ noise amplification per stage [7]) and, more fundamentally, by the basis's spatial resolution – once skin depth $\delta(\omega) = \sqrt{2/(\omega\mu\sigma)}$ drops below the basis length scale, further rungs are non-physical. In that regime, however, the Surface Impedance Boundary Condition $Z_s(j\omega) = \sqrt{j\omega\mu/\sigma}$ – a Constant Phase Element of exponent $\alpha = 1/2$ – becomes exact.

We seam the two representations at a turning point ω_N given by Matsuo's corner-frequency formula [3] evaluated at the last reliable rung,

$$\omega_N = \sqrt{2^{\alpha/2} - 1}\, R_{2N-2}/L_{2N-1} \approx 0.4350\, R_{2N-2}/L_{2N-1}, \quad (1)$$

yielding $Y_{unified}(j\omega) = 1/Z^{(N)}_{Cauer}(j\omega)$ for $\omega < \omega_N$ and $K_{SIBC}\sqrt{\sigma/(j\omega\mu)}$ above, with $K_{SIBC} = Y^{(N)}_{Cauer}(j\omega_N)/\sqrt{\sigma/(j\omega_N\mu)}$ fixed by continuity (no fitting). This extension is the s_∞ expansion point of the multiple-expansion-points CLN framework [2]: $K_\infty/s_\infty \to \sigma$ degenerates the energy operator to the conductivity inner product, the natural basis for surface-localised modes.

II. Verification

Slab. For a Cu slab of half-thickness $c = 1\,\text{mm}$, Stoll's $Y(s) = \sigma c \tanh(u)/u$, $u = c\sqrt{s\mu\sigma}$ [4] with Wall's odd-integer continued fraction [5] gives the analytical ladder. Our mpmath QD-Wynn extractor at 60-digit precision recovers ten Cauer rungs; under the plate-symmetry constraint $k_L = k_R$ [3] the asymptotic fit returns $\alpha = 0.500$ (Warburg). ω_N at $N = 10$ gives $f_N = 1.12\,\text{MHz}$, $K_{SIBC} = 0.937$, and $Y_{unified}$ matches Y_{full} to under 10 % at the seam and 1 % deep in either regime from DC to 10^8 Hz.

3D Cu cuboid (17.72×17.72×2 mm). A pure-mpmath single-element Volume Integral Method (VIM) with HDiv divergence-free hex basis [6] at $p = 3$ (81 dof) yields two reliable Cauer rungs $\tau_{pair} = 194.66, 69.96\,\mu\text{s}$ and a third $19.40\,\mu\text{s}$ that lies in the basis noise floor; the *mesh-natural* truncation seams the SIBC element at $f_N = 9.9 \times 10^2$ Hz. An NGSolve HCurl Tanimoto-AT extraction produces an extra leading stage corresponding to the $t = 0^+$ skin-layer impulse response; identifying it as the SIBC seed and treating stages $\{1, 2, 3\}$ as the bulk Cauer ladder yields three rungs $\tau_{pair} = 218.99, 84.16, 58.03\,\mu\text{s}$ that are *bit-identical* across $h = 0.4$–0.5 mm at $p = 3$ and $p = 3$–4 at fixed mesh, confirming mesh and order convergence. The third FEM rung $L_5 = 9.43 \times 10^{-8}$ provides the seam $f_N = 1.19\,\text{kHz}$; the first two FEM rungs agree with VIM within 12–20 %. The unified $Y(j\omega)$ from VIM (two rungs, $f_N = 990\,\text{Hz}$) and FEM (three rungs, $f_N = 1190\,\text{Hz}$) overlap to graphical precision (Fig. 1). A direct NGSolve frequency-domain AC sweep ($A_{red} \cdot t = 0$ Dirichlet, 20 frequencies 1 Hz–10^7 Hz) provides the ground-truth $Y_{AC}(j\omega) = |A_{ext}|^2 + \langle A_{red}, A_{ext}\rangle$, which coincides with the unified prediction throughout the Cauer regime (1 kHz) and decays faster than the SIBC seam above it, consistent with the finite-conductor Faraday-shielding floor.

References

[1] A. Kameari *et al.*, IEEE Trans. Magn. **54**, 3, 2018.
[2] K. Kuriyama *et al.*, IEEE Trans. Magn. **55**, 6, 2019.
[3] T. Matsuo, IEEJ SA/RM-26-014, 2026.
[4] R. L. Stoll, *The Analysis of Eddy Currents*, Clarendon, 1974.
[5] H. S. Wall, *Analytic Theory of Continued Fractions*, Van Nostrand, 1948.
[6] J. Schöberl and S. Zaglmayr, COMPEL **24**, 374, 2005.
[7] M. Nagamine *et al.*, submitted, 2026.

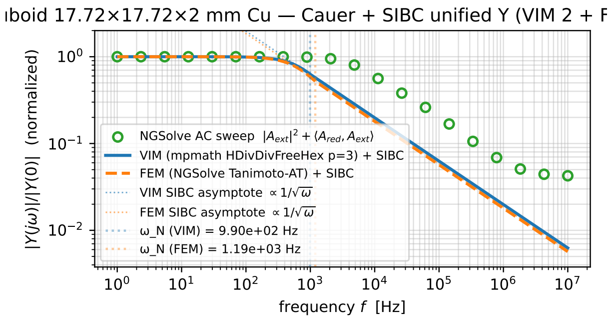

Fig. 1. Cauer + SIBC unified $|Y(j\omega)|/|Y(0)|$ for the 17.72×17.72×2 mm Cu cuboid in B_z. VIM (solid, two Cauer rungs at $p = 3$, 81 dof) and NGSolve Tanimoto-AT (dashed, three mesh-converged rungs at $p = 3$, $h = 0.5$ mm) overlap across nine decades. Green circles: independent NGSolve AC sweep ground truth – matches the unified prediction in the Cauer regime and lies slightly below the SIBC seam at high ω owing to finite-conductor Faraday shielding.